

3.7 Summary and Notes

3.7.1 Summary

Idea.

1. $\mathbf{Y}$ is multivariate normal when every linear combination $\mathbf{a}^T \mathbf{Y}$ is normal, constants included (Definition 3.1.1). The definition covers singular covariance matrices, which regression needs for fitted values and residuals.
2. $\boldsymbol{\mu} + \mathbf{A}\mathbb{Z}$, with $\mathbb{Z}$ standard normal, is $N(\boldsymbol{\mu}, \mathbf{A}\mathbf{A}^T)$. Every mean and every nonnegative definite covariance occurs (Theorem 3.1.2).
3. The moment generating function is $\exp(\mathbf{t}^T \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})$, so a normal distribution is determined by $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ (Theorem 3.2.1). When $\boldsymbol{\Sigma}$ is positive definite there is a density whose contours are the ellipsoids of constant Mahalanobis distance, and $\Delta^2(\mathbf{Y}) \sim \chi^2(n)$ (Theorem 3.2.2 and Proposition 3.2.3).
4. Affine maps of normal vectors are normal, with no rank condition, and so are all subvectors (Theorem 3.3.1 and Corollary 3.3.2). A normal vector of rank r is an exact affine image of r independent standard normals (Proposition 3.3.4).
5. Normal margins do not imply joint normality, and zero correlation between normal variables does not imply independence without it (Example (A random sign) and Example (A mixture of two correlations)).
6. For jointly normal blocks, zero cross-covariance is equivalent to independence. $\mathbf{A}\mathbf{Y}$ and $\mathbf{B}\mathbf{Y}$ are independent iff $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^T = \mathbf{0}$ (Theorem 3.4.1 and Corollary 3.4.2).
7. $\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{y}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11.2})$: linear mean, constant covariance, reduced variance (Theorem 3.5.2). The conditional mean is the best predictor of all, and coincides with the best linear predictor (Proposition 3.5.3).
8. The partial correlation is the correlation of the residuals from linear prediction on the conditioning variables. Under normality it is the conditional correlation, and it is zero iff the variables are conditionally independent (Definition 3.6.1 and Proposition 3.6.1). It can be computed by a recursion, from the inverse covariance matrix, or by FWL residuals (Proposition 3.6.2 and Proposition 3.6.3).

3.7.2 Notes and sources

Three definitions. Linear-model texts reach the multivariate normal by different routes, and comparing them explains the choices made here. Rencher and Schaalje (2008, chapter 4) start from the positive definite case, obtain the density by transforming independent standard normals with $\boldsymbol{\Sigma}^{1/2}$, and prove the properties of linear functions and independence with moment generating functions, for full-rank transformations. Seber and Lee (2003, chapter 2) also begin with the density, then extend the definition to singular covariance matrices through the representation $\mathbf{A}\mathbb{Z} + \boldsymbol{\mu}$ and prove that normality of all linear combinations characterizes the family. Their moment-generating-function proof of the independence theorem is the model for [Section 3.4](#). Christensen (2020, §1.2) takes $\mathbf{A}\mathbb{Z} + \boldsymbol{\mu}$ as the definition from the outset, shows with characteristic functions that the law depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, and proves the independence theorem by building an explicit independent version. Defining the distribution through linear combinations, as this chapter does, goes back at least to Rao (1973, chapter 8). It has the

advantage that the theorem on affine transformations needs no rank condition and a one-line proof.

Cramér–Wold. The device of reducing a vector to its linear combinations is from Cramér and Wold (1936), where it was used to reduce multivariate limit theorems to univariate ones. [Chapter 2](#) proves it, together with the uniqueness and factorization properties of moment generating functions used in [Section 3.2](#). Billingsley (1995) is the reference for these facts and for the conditioning property of independent variables used in the proof of Theorem 3.5.2.

Conditional distributions and partial correlation. The proof of Theorem 3.5.2 through the independent error $\mathbf{W}$ borrows one idea from Rencher and Schaalje (2008) and Seber and Lee (2003): both pass through the transformed vector $\mathbf{Y}_1 - \mathbf{B}\mathbf{Y}_2$, which is independent of $\mathbf{Y}_2$, although both still obtain the conditional density as a ratio of densities in the positive definite case. The proof here uses only independence and a generalized inverse, so it covers singular Σ_{22} , an extension that is standard in multivariate analysis (Rao 1973). Anderson (2003) is the classical reference for the multivariate normal in statistics, including the distribution theory of sample partial and multiple correlation coefficients that Chapter 14 uses. Rencher and Schaalje (2008, §4.5) define partial correlation as a conditional correlation, as in Proposition 3.6.1(b). The residual interpretation (a) does not require normality and is the one that connects to least squares. Regression toward the mean was described by Galton (1886). The link between zeros of the inverse covariance matrix and conditional independence was made the basis of a modelling strategy by Dempster (1972), and Lauritzen (1996) develops the theory of graphical models that grew from it.

Data. Example (Employment and unemployment) uses the macroeconomic series published by Longley (1967) as a test of numerical accuracy, available in `statsmodels`. The same series appear in [Chapter 2](#), as an example of sample moments, and in [Chapter 6](#), to illustrate ill-conditioning.

3.7.3 References

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